# Quantitative Literacy:

## Why a Historian Teaches Quantitative Analysis

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## 1 Quantitative Literacy

I have always been cautious about quantitative methods. Not because I dislike numbers, but because I distrust the authority that numbers are often given. Statistics can illuminate patterns, but they can also conceal complexity. Indicators can simplify reality, yet what they leave out is often just as important as what they measure. Jerry Z. Muller's *The Tyranny of Metrics* captures this concern well. When measurement becomes an end in itself, we risk valuing only what can be counted while overlooking what truly matters.

For this reason, I do not believe that quantitative analysis should ever stand alone as the sole basis for an argument. As Theodore Porter and many others have argued, "numbers do not speak for themselves'. Every dataset reflects choices about what to collect, what to ignore, how to classify observations, and how to interpret results. A statistical model is not a mirror of reality but one possible representation of it. Without historical context, qualitative evidence, and careful interpretation, quantitative findings can easily be misunderstood or misused. Yet my skepticism does not lead me to reject quantitative methods. Quite the opposite.

We live in an age in which governments, journalists, consultants, and researchers increasingly justify their claims by appealing to "the evidence."[1] Public policies are expected to be evidence-based. Reports are filled with charts and statistical models. Social media circulates colorful visualizations that appear objective and persuasive. Whether we like it or not, quantitative evidence now plays a central role in public debate.

Quantitative literacy requires more than statistical skills; it requires access to data. In Japan, government statistics are often published not as raw datasets but as pre-aggregated tables. As a result, citizens and researchers have little opportunity to reproduce analyses or to examine whether the government's interpretation of the numbers is the only plausible one. Too often, the process through which evidence is transformed into official conclusions and policy recommendations remains opaque. Without access to the underlying data and methods, that process becomes a black box rather than an object of public scrutiny.

For that very reason, we should approach such evidence with the same critical spirit that Samuel Johnson once brought to political eloquence. Johnson urged his readers to ask whether they were persuaded by

[1] In contemporary Japan, dismissing an argument as "lacking evidence" has itself become a powerful rhetorical strategy. Yet those who make such claims do not always make their own evidence, data, or methods equally open to public scrutiny.

the truth of an argument or merely by the ornament of its rhetoric. Today, we should ask an analogous question: are we persuaded by the quality of the evidence itself, or by the elegance of its visualization? Do we judge the evidence on its own merits, or do we place undue trust in the institutions and experts who present it? Answering those questions requires not blind faith in data or in their interpretation, but a basic understanding of how quantitative evidence is produced, analyzed, and presented.

This is precisely why I believe students in the humanities and social sciences should learn quantitative methods. My goal is not to turn every student into a statistician or a data scientist. I do not present myself as a data scientist, nor do I aspire to become one. I teach quantitative methods from the perspective of a historian.

When someone claims that "the data show" or "research proves," students should be able to ask basic but important questions. What data were collected? How were the variables defined? Which observations were excluded? What assumptions underlie the statistical model? Does the visualization faithfully represent the data, or does it exaggerate a particular conclusion? These questions do not require advanced mathematics. They require curiosity, skepticism, and a willingness to look beyond the surface of a graph or a table.

I also believe that learning quantitative methods can enrich, rather than replace, qualitative inquiry. A graph may reveal an unexpected pattern, but it cannot explain why that pattern exists. Statistical analysis can identify relationships, but it cannot fully capture historical contingency, political ideas, or human experience. Quantitative and qualitative approaches answer different kinds of questions, and they are most powerful when they inform one another.

Ultimately, I believe quantitative methods deserve a place in the education of students in the humanities and social sciences—not because numbers reveal the truth, but because numbers are increasingly used to claim the truth, while many accept such claims without recognizing the assumptions and judgments that lie behind them. Students should understand how quantitative evidence is produced, analyzed, and visualized—not so that they accept it uncritically, but so that they can evaluate it thoughtfully. In a society where decisions are increasingly justified by data, quantitative literacy is not merely a technical skill. It is an essential part of democratic citizenship and responsible scholarship. This conviction is especially important in Japan, where discussions of data science education have been led primarily by technical disciplines rather than by the humanities and social sciences[2].

## 2 The R Community

I teach with R, a free and open-source programming language for statistical computing, and have increasingly incorporated it into my own research. My preference for R is rooted in the same commitment to democratic access to quantitative literacy. The tidyverse, a widely used collection of R packages led by Hadley Wickham, embodies that philosophy by making data analysis more accessible, more consistent, and easier to share. The tidyverse is not simply a collection of packages; it is an attempt to lower the barriers to quantitative analysis without lowering intellectual standards.

[2]While information engineering and computer science have made indispensable contributions to data science education in Japan, discussions have tended to emphasize technical competence over the broader civic and interpretive dimensions of quantitative literacy. My aim is not to challenge this expertise but to argue that historians and other scholars in the humanities and social sciences also have an important role to play in shaping quantitative education.

As Wickham himself recounts in his personal history of the tidyverse, what began as an individual project has become a collaborative effort sustained by a vibrant global community. Beyond the software itself, I value the culture of the R community. Thousands of volunteers devote their time to developing, improving, and freely sharing packages because they believe that what they have built may be useful to others. This spirit of openness and collaboration embodies the idea that knowledge advances not only through competition, but also through generosity[3]. It is also one of the reasons I choose to teach quantitative analysis with R.

For anyone interested in the philosophy behind the tidyverse, I highly recommend Hadley Wickham's essay, 'A personal history of the tidyverse'.

[3]Wes McKinney, a key figure in the open-source data science ecosystem, emphasizes that building open-source projects requires not only technical innovation but also sustainable communities and support structures. Reflecting on his decision to join Posit, he describes the company's mission as "open source data science for public good" and highlights the importance of "technical excellence, sustainable growth, and supporting the public good." His discussion illustrates a broader challenge for open-source communities: how to preserve openness and collaboration while creating the conditions necessary for long-term sustainability. See Wes McKinney, 'Joining Posit's Polyglot Data Science Mission'.